Critical Gaussian sparsity need not converge to the semicircle edge

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Abstract

At the critical degree $k \sim a \log n$, Bandeira and van Handel ask whether the norm of every Gaussian matrix supported on a k-regular graph might converge, after division by $\sqrt{k}$, to the semicircle edge 2 when a is sufficiently large. We give a negative answer to this universal alternative for every finite $a > 0$. Take the support graph to be a disjoint union of copies of $K_{k,k}$. The random matrix becomes a direct sum of independent Gaussian bipartite blocks, whose norms are largest singular values of square Gaussian matrices. A standard Wishart right-tail large deviation principle and an elementary extreme-value argument show that

$$\frac{\|X\|}{\sqrt{k}} \longrightarrow t_a > 2, \qquad \int_2^{t_a} \sqrt{x^2-4}\,dx = \frac{1}{a}.$$

Thus an arbitrarily large fixed logarithmic-degree constant still permits a strict outlier above the bulk.

1 The source question

The source considers a symmetric matrix $X = (b_{ij} g_{ij})_{i,j \le n}$, where the independent upper-triangular variables are standard Gaussian and $B = (b_{ij})$ is a symmetric $\{0,1\}$ matrix having exactly k nonzero entries in every row. Corollary 4.4 proves $\|X\|/\sqrt{k} \to 2$ when $k/\log n \to \infty$, divergence when $k/\log n \to 0$, and boundedness at $k \sim \log n$. Remark 4.5 asks whether at $k = a \log n$ separation from 2 persists for all a, or instead the norm converges to 2 for sufficiently large a [1].

$N(0,1)$ random variables which is of order $\sqrt{2\log(n/2)}$ as $n \to \infty$. Thus $\|X\|/\sqrt{k}$ diverges.

For $k/\log n \to \infty$, we note that Corollary 3.9 yields

$$\mathbf{P}[\|X\|/\sqrt{k} \ge 2+\varepsilon] \le n e^{-C_\varepsilon k} = n^{1-C_\varepsilon k/\log n}$$

for a suitable constant C_ε. Thus $\|X\|/\sqrt{k} \le 2+\varepsilon+o(1)$ for any $\varepsilon > 0$. On the other hand, Theorem 4.3 implies that $\|X\|/\sqrt{k} \ge 2-\varepsilon-o(1)$ for any $\varepsilon > 0$.

If $k = a\log n$, Corollary 3.9 similarly yields that $\mathbf{P}[\|X\|/\sqrt{k} > C] \to 0$ for a sufficiently large constant C, so $\{\|X\|/\sqrt{k}\}$ is bounded. However, Lemma 3.14 shows that $\mathbf{E}\|X\|/\sqrt{k} \gtrsim a^{-1/2} > 3$ when a is sufficiently small. $\square$

Remark 4.5. We have not investigated the precise behavior of $\|X\|/\sqrt{k}$ for the boundary case $k = a\log n$. The proof of Corollary 4.4 shows that if a is chosen sufficiently small, the rescaled norm remains bounded but strictly separated from the bulk as $n \to \infty$. We do not know whether this is the

26 A. S. BANDEIRA AND R. VAN HANDEL

case for all a, or whether the norm does in fact converge to 2 when a is sufficiently large. A precise investigation of this question is beyond the scope

The hypotheses allow the support graph to depend arbitrarily on n. We disprove the possibility of a universal large-a convergence theorem under these hypotheses. The stronger classification of which support graphs do or do not converge to 2 remains outside the claim.

2 Result

For $t \geq 2$, put

$$J(t) := \int_2^t \sqrt{x^2 - 4}\, dx = \frac{1}{2}\left(t\sqrt{t^2-4} - 4\,\mathrm{arcosh}(t/2)\right). \tag{1}$$

The function J is continuous and strictly increasing on $[2, \infty)$, with $J(2) = 0$ and $J(t) \to \infty$.

Theorem 1. *Fix $a > 0$. There are integers $n_k \to \infty$, symmetric $\{0,1\}$ matrices B_k of size n_k, and Gaussian matrices $X_k = ((B_k)_{ij} g_{ij})$ of the type considered in [1], such that every row of B_k has exactly k nonzero entries,*

$$\frac{k}{\log n_k} \longrightarrow a,$$

and

$$\frac{\|X_k\|}{\sqrt{k}} \xrightarrow{\ \mathbf{P}\ } t_a, \qquad J(t_a) = \frac{1}{a}. \tag{2}$$

In particular $t_a > 2$ for every finite $a > 0$.

The source's bulk conclusion still applies: $k = o(n_k)$, so the empirical spectral distribution converges to the semicircle law while the extreme eigenvalue remains separated from its edge.

3 Proof

We isolate the only probabilistic input. Let G_k be a $k \times k$ matrix with independent $N(0,1)$ entries and let $S_k = \|G_k\|$ be its largest singular value.

Lemma 2 (square-Wishart right tail)**.** *For every $t > 2$,*

$$\lim_{k\to\infty} \frac{1}{k} \log \mathbf{P}\left\{\frac{S_k}{\sqrt{k}} \geq t\right\} = -J(t). \tag{3}$$

Proof. This is the square real-Gaussian case of the Wishart large deviation principle. For completeness, we spell out the normalization from [2, Theorem 1.7]. The largest eigenvalue of

$$\frac{1}{\sqrt{2k}}\begin{pmatrix} 0 & G_k \\ G_k^* & 0 \end{pmatrix}$$

is $S_k/\sqrt{2k}$. At aspect ratio one, the theorem gives speed $2k$, edge $\sqrt{2}$, and rate

$$I(x) = \int_{\sqrt{2}}^x \sqrt{y^2 - 2}\, dy, \qquad x \geq \sqrt{2}.$$

As the rate is continuous above the edge, applying the LDP to the tail and substituting $x = t/\sqrt{2}$ gives

$$2I(t/\sqrt{2}) = \int_2^t \sqrt{u^2 - 4}\, du = J(t),$$

which is (3). □

Proof of Theorem 1. Let

$$m_k = \lfloor e^{k/a} \rfloor, \qquad n_k = 2km_k,$$

and let the support graph of B_k be the disjoint union of m_k copies of $K_{k,k}$. Every vertex has degree exactly k, and

$$\log n_k = \frac{k}{a} + O(\log k), \qquad \frac{k}{\log n_k} \longrightarrow a.$$

After grouping coordinates by components, the associated Gaussian matrix is a direct sum

$$X_k = \bigoplus_{r=1}^{m_k} \begin{pmatrix} 0 & G_{k,r} \\ G_{k,r}^* & 0 \end{pmatrix},$$

where the $G_{k,r}$ are independent $k \times k$ standard Gaussian matrices. Consequently

$$\frac{\|X_k\|}{\sqrt{k}} = \max_{1 \le r \le m_k} \frac{\|G_{k,r}\|}{\sqrt{k}}. \tag{4}$$

There is a unique $t_a > 2$ satisfying $J(t_a) = 1/a$. Fix $0 < \varepsilon < t_a - 2$ and abbreviate

$$p_k(t) = \mathbf{P}\{S_k/\sqrt{k} \ge t\}.$$

By Lemma 2,

$$\begin{aligned} \log(m_k p_k(t_a - \varepsilon)) &= k\{a^{-1} - J(t_a - \varepsilon) + o(1)\} \longrightarrow +\infty, \\ \log(m_k p_k(t_a + \varepsilon)) &= k\{a^{-1} - J(t_a + \varepsilon) + o(1)\} \longrightarrow -\infty. \end{aligned}$$

Independence and $1 - x \le e^{-x}$ therefore give

$$\mathbf{P}\left\{\max_{r \le m_k} \frac{\|G_{k,r}\|}{\sqrt{k}} < t_a - \varepsilon\right\} = (1 - p_k(t_a - \varepsilon))^{m_k} \le e^{-m_k p_k(t_a - \varepsilon)} \longrightarrow 0,$$

while the union bound gives

$$\mathbf{P}\left\{\max_{r \le m_k} \frac{\|G_{k,r}\|}{\sqrt{k}} \ge t_a + \varepsilon\right\} \le m_k p_k(t_a + \varepsilon) \longrightarrow 0.$$

Together with (4), this proves (2). □

4 Consequences and scope

Since $J(t) = \frac{4}{3}(t-2)^{3/2}(1 + o(1))$ as $t \downarrow 2$,

$$t_a = 2 + \left(\frac{3}{4a}\right)^{2/3} (1 + o(1)) \qquad (a \to \infty). \tag{5}$$

Thus the gap becomes small for large a, but it is strictly positive for every fixed finite a. The mechanism is structural: each individual Gaussian block has norm near $2\sqrt{k}$, but among $m_k = \exp(k/a + o(k))$ independent blocks a speed-k upper-tail deviation occurs with high probability.

This settles the universal interpretation of the second alternative in Remark 4.5: no finite threshold in a can force convergence to 2 for all regular support patterns. It does not assert separation for every k-regular graph; expansion or other global structure may produce different critical behavior.

The distinct Gaussian row-norm question in Remark 3.16 of the source is not newly solved here: it was subsequently answered by Latała, van Handel, and Youssef [3]. A bounded search on 21 July 2026 found the Wishart LDP and later Gaussian norm results, but no source explicitly drawing the $K_{k,k}$ critical-block consequence proved above. This is a novelty check, not a definitive certification.

References

[1] A. S. Bandeira and R. van Handel, *Sharp nonasymptotic bounds on the norm of random matrices with independent entries*, Ann. Probab. 44 (2016), 2479–2506; https://arxiv.org/abs/1408.6185.

[2] A. Guionnet and J. Husson, *Large deviations for the largest eigenvalue of Rademacher matrices*, Ann. Probab. 48 (2020), 1436–1465; https://arxiv.org/abs/1810.01188.

[3] R. Latała, R. van Handel, and P. Youssef, *The dimension-free structure of nonhomogeneous random matrices*, Invent. Math. 214 (2018), 1031–1080; https://arxiv.org/abs/1711.00807.